ASSIGNMENT- 01

Bachelor of Computer Applications (Hons. AI & ML)
Semester- III

Subject: Database Management System

Topic: Identifying Tables for a Real-World Application

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August 2026

1. Introduction

Database creation and table implementation are important steps in developing a database management system. This assignment implements the Paranormal Investigation Agency Management System using SQL Data Definition Language (DDL).

The database stores information about clients, investigators, locations, case categories, equipment, investigations, evidence, reports, and payments. Primary keys and foreign keys are used to identify records and establish relationships between tables.

2. Objective

- To create a database for a Paranormal Investigation Agency Management System.
- To create all 11 tables using SQL DDL commands.
- To define primary keys and foreign keys.
- To establish relationships between the tables.
- To implement the database structure according to Assignment 1.

3. Software Used

DBMS: MySQL
Language: SQL
Commands Used: `CREATE DATABASE`, `USE`, and `CREATE TABLE`

4. SQL Implementation

4.1 Create the Database

CREATE DATABASE Paranormal_Investigation_Agency;

USE Paranormal_Investigation_Agency;

Explanation:
The `CREATE DATABASE` command creates a new database named `Paranormal_Investigation_Agency`. The `USE` command selects the database in which the tables will be created.

4.2 Create the Clients Table

```
CREATE TABLE Clients (
    client_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100),
    phone VARCHAR(20),
    address VARCHAR(255),
    registration_date DATE
);
```

Explanation:
The Clients table stores the personal and contact details of people who request paranormal investigations. The `client_id` uniquely identifies each client.

4.3 Create the Investigators Table

```
CREATE TABLE Investigators (
    investigation_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    speciliazation VARCHAR(100),
    phone VARCHAR(20),
    email VARCHAR(100),
    experience_level VARCHAR(50),
    date_joined DATE
);
```

Explanation:
The Investigators table stores information about agency staff. The `investigation_id` is the primary key used to identify each investigator, following the column name in Assignment 1.

4.4 Create the Locations Table

```
CREATE TABLE Locations (
   location_id INT PRIMARY KEY AUTO_INCREMENT,
   addess VARCHAR(255) NOT NULL,
   location_type VARCHAR(100),
   year_built INT,
   known_history TEXT
);
```

Explanation:
The Locations table stores information about properties and sites being investigated. A location can be associated with multiple cases over time.

4.5 Create the Case_Categories Table

```
CREATE TABLE Case_Categories (
   Category_id INT PRIMARY KEY AUTO_INCREMENT,
   category_name VARCHAR(100) NOT NULL,
   description TEXT
);
```

Explanation:
The Case_Categories table classifies investigations into categories such as Residential Haunting, Poltergeist Activity, EVP Session, Demonic Activity, and Debunking Request.

4.6 Create the Equipment Table

```
CREATE TABLE Equipment (
   equipment_id INT PRIMARY KEY AUTO_INCREMENT,
   equipment_name VARCHAR(100) NOT NULL,
```

```
    equipment_type VARCHAR(50),
    condition_status VARCHAR(50),
    purchase_date DATE
);
```

Explanation:
The Equipment table maintains an inventory of devices used during investigations, including EMF meters, cameras, and digital voice recorders.

4.7 Create the Cases Table

```
CREATE TABLE Cases (
    Case_id INT PRIMARY KEY AUTO_INCREMENT,
    Client_id INT NOT NULL,
    Location_id INT NOT NULL,
    Category_id INT NOT NULL,
    Date_reported DATE NOT NULL,
    Investigation_date DATE,
    Case_status VARCHAR(50),
    Urgency_level VARCHAR(20),

    FOREIGN KEY (Client_id)
        REFERENCES Clients(client_id),

    FOREIGN KEY (Location_id)
        REFERENCES Locations(location_id),

    FOREIGN KEY (Category_id)
```

```
    REFERENCES Case_Categories(Category_id)
);
```

Explanation:
The Cases table is the central table of the system. It records each investigation and connects it to the relevant client, location, and case category through foreign keys.

4.8 Create the Case_Investigator Table

```
CREATE TABLE Case_Investigator (
  Case_id INT,
  Investigator_id INT,
  Role_in_case VARCHAR(50),

  PRIMARY KEY (Case_id, Investigator_id),

  FOREIGN KEY (Case_id)
    REFERENCES Cases(Case_id),

  FOREIGN KEY (Investigator_id)
    REFERENCES Investigators(investigation_id)
);
```

Explanation:
The Case_Investigator table resolves the many-to-many relationship between cases and investigators. A case can have multiple investigators, and an investigator can work on multiple cases. The composite primary key prevents the same investigator from being assigned to the same case more than once.

4.9 Create the Case_Equipment Table

```
CREATE TABLE Case_Equipment (
    Case_id INT,
    Equipment_id INT,
    Checkout_time DATETIME,
    Return_time DATETIME,

    PRIMARY KEY (Case_id, Equipment_id, Checkout_time),

    FOREIGN KEY (Case_id)
        REFERENCES Cases(Case_id),

    FOREIGN KEY (Equipment_id)
        REFERENCES Equipment(equipment_id)
);
```

Explanation:
The Case_Equipment table records the equipment assigned to each case, including checkout and return times. Its composite primary key allows the same equipment to be recorded for different checkout events.

4.10 Create the Evidence Table

```
CREATE TABLE Evidence (
    Evidence_id INT PRIMARY KEY AUTO_INCREMENT,
    Case_id INT NOT NULL,
    Recorded_by INT NOT NULL,
    Evidence_type VARCHAR(100),
    Description TEXT,
    File_refernce VARCHAR(255),
```

```
    timestamp DATETIME,

    FOREIGN KEY (Case_id)
        REFERENCES Cases(Case_id),

    FOREIGN KEY (Recorded_by)
        REFERENCES Investigators(investigation_id)
);
```

Explanation:
The Evidence table records findings collected during an investigation, such as photographs, audio recordings, videos, EMF spikes, and eyewitness observations. Each evidence record is linked to a case and the investigator who recorded it.

4.11 Create the Reports Table

```
CREATE TABLE Reports (
    Report_id INT PRIMARY KEY AUTO_INCREMENT,
    Case_id INT NOT NULL UNIQUE,
    Prepared_by INT NOT NULL,
    Final_verdict VARCHAR(50),
    Summary TEXT,
    Report_date DATE,

    FOREIGN KEY (Case_id)
        REFERENCES Cases(Case_id),

    FOREIGN KEY (Prepared_by)
        REFERENCES Investigators(investigation_id)
```

);

Explanation:
The Reports table stores the final findings and conclusions of investigations. The `UNIQUE` constraint on `Case_id` ensures that each case has at most one report, consistent with the one-to-one relationship in Assignment 1.

4.12 Create the Payments Table

```
CREATE TABLE Payments (
    Payment_id INT PRIMARY KEY AUTO_INCREMENT,
    Case_id INT NOT NULL UNIQUE,
    Amount DECIMAL(10,2),
    Payment_mode VARCHAR(50),
    Payment_status VARCHAR(50),
    Payment_date DATE,

    FOREIGN KEY (Case_id)
        REFERENCES Cases(Case_id)
);
```

Explanation:
The Payments table stores billing information for investigation services, including the amount charged, payment method, payment status, and payment date. The `UNIQUE` constraint on `Case_id` ensures that each case is associated with at most one payment record, following the original design.

5. Primary Keys and Foreign Keys

Primary Keys

A primary key uniquely identifies each record in a table.

Table	Primary Key
Clients	client_id
Investigators	investigation_id
Locations	location_id
Case_Categories	Category_id
Equipment	equipment_id
Cases	Case_id
Case_Investigator	Case_id, Investigator_id
Case_Equipment	Case_id, Equipment_id, Checkout_time
Evidence	Evidence_id
Reports	Report_id
Payments	Payment_id

Foreign Keys

Foreign keys connect related tables and maintain referential integrity.

Table	Foreign Key	References
Cases	Client_id	Clients(client_id)
Cases	Location_id	Locations(location_id)
Cases	Category_id	Case_Categories(Category_id)
Case_Investigator	Case_id	Cases(Case_id)
Case_Investigator	Investigator_id	Investigators(investigation_id)
Case_Equipment	Case_id	Cases(Case_id)
Case_Equipment	Equipment_id	Equipment(equipment_id)
Evidence	Case_id	Cases(Case_id)
Evidence	Recorded_by	Investigators(investigation_id)
Reports	Case_id	Cases(Case_id)
Reports	Prepared_by	Investigators(investigation_id)
Payments	Case_id	Cases(Case_id)

6. Verify the Tables

After executing all the SQL commands, use the following command to display the tables created in the database.

SHOW TABLES;

Expected Output:

The output should list the following 11 tables:

1. **Clients**
2. **Investigators**
3. **Locations**
4. **Case_Categories**
5. **Equipment**
6. **Cases**
7. **Case_Investigator**
8. **Case_Equipment**
9. **Evidence**
10. **Reports**
11. **Payments**

To inspect the structure of a table, execute:

DESCRIBE Cases;

Explanation:
The `SHOW TABLES` command displays all tables in the selected database. The `DESCRIBE` command displays a table's columns, data types, key information, and nullability.

7. Result

The `Paranormal_Investigation_Agency` database was designed using SQL DDL commands. All 11 tables were defined with suitable data types, primary keys, and foreign keys. The relationships between clients, investigators, locations, cases, equipment, evidence, reports, and payments were implemented according to the proposed database design.

8. Conclusion

This assignment demonstrated the creation of a relational database for a Paranormal Investigation Agency Management System. The database structure organises information into separate tables and connects related records using primary and foreign keys. This provides a foundation for inserting sample data, applying integrity constraints, and executing SQL queries in the following assignments.